Kinematic Analysis and Explainability of the S-JEPA Architecture on the NTU RGB+D 120 Dataset

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***Abstract*—This study presents a comprehensive Data Science pipeline applied to the 3D skeleton-based action recognition task on the NTU RGB+D 120 dataset. The pipeline begins with qualitative and quantitative data classification, incorporating exploratory data analysis (EDA) via line and bar charts. In the preprocessing phase, the data is rigorously evaluated using six quality criteria, coupled with cleaning techniques such as Tukey's statistical outlier filtering, DBSCAN density clustering, and data normalization. To construct view-invariant features, we verify independence using the Chi-square test and analyze feature correlations using Pearson and Spearman coefficients. The spatio-temporal kinematic features are then self-supervisedly pre-trained via the S-JEPA (Spatio-temporal Joint-Embedding Predictive Architecture) model. Finally, we apply feature selection techniques (Filter, Wrapper, Embedded), dimensionality reduction transformations (PCA, Kernel PCA, ISOMAP, LDA), and eXplainable AI (XAI) methods including SHAP, Attention Rollout, and Gradient Saliency (the foundation of Grad-CAM and Integrated Gradients) to demystify the learned latent feature space. Experimental results demonstrate that the proposed multi-stream S-JEPA framework achieves competitive performance, reaching 78.30% on NTU-120 X-Sub and 80.50% on NTU-120 X-Set through weighted late fusion, while maintaining high computational efficiency.**

***Index Terms*—Data Science Pipeline, NTU RGB+D 120, Data Preprocessing, Correlation Analysis, Dimensionality Reduction, S-JEPA, Explainable AI, SHAP, Gradient Saliency.**

I. Introduction

Human action recognition is fundamental to numerous intelligent applications, including security surveillance, human-computer interaction, and medical behavioral analysis [1], [11]. Among these, 3D skeleton data is preferred due to its robustness against environmental noise and computational efficiency.

This study implements a standard data science pipeline consisting of: Collection → Processing → Cleaning → EDA → Modeling → Reporting/Visualization → Decision Making. To overcome the barrier of high-quality label scarcity, we train the next-generation self-supervised learning model S-JEPA (Joint-Embedding Predictive Architecture). We focus on data normalization, outlier removal, feature correlation analysis between joints, and decoding the decision-making mechanism of the deep learning model through advanced XAI methods to enhance the reliability and transparency of the system.

II. Related Work

A. Self-supervised Learning for Skeleton-based Action Recognition

Self-supervised learning (SSL) methods for 3D skeleton data are primarily categorized into three main groups:

- **Generative Architectures:** Typically SkeletonMAE [4] and MAMP [3], which are based on a masking mechanism and reconstruct physical joint coordinates or temporal kinematics in the input data space. These models are susceptible to high-frequency noise from raw coordinates.
- **Contrastive Learning:** Includes studies like AimCLR [5] and ActCLR [6], focusing on maximizing the similarity between different augmentations of the same action sequence.
- **Joint-Embedding Predictive Architecture:** S-JEPA [7] learns to match the probability distribution between predicted and target representations via an EMA mechanism, avoiding representation collapse without the need for negative contrastive samples.

B. Explainable AI (XAI)

To overcome the "black box" barrier of deep neural networks, two main branches of XAI techniques have been developed:

- **Model-Agnostic Techniques:** Include LIME [10] (local linear approximation), ANCHORS (If-Then decision rules), and SHAP [10] (Shapley marginal contribution allocation based on cooperative game theory).
- **Vision Network Interpretation (CNN/Transformer-specific):** Include CAM, Grad-CAM [9] (utilizing gradients of the final convolutional layer), Grad-CAM++ (improved multi-object localization), and Integrated Gradients (IG) (gradient integral along the path from a baseline image).

III. Data Science Pipeline and Exploratory Data Analysis (EDA)

We implement the exploratory data analysis (EDA) phase to statistically analyze and identify characteristic properties within the NTU RGB+D 120 dataset.

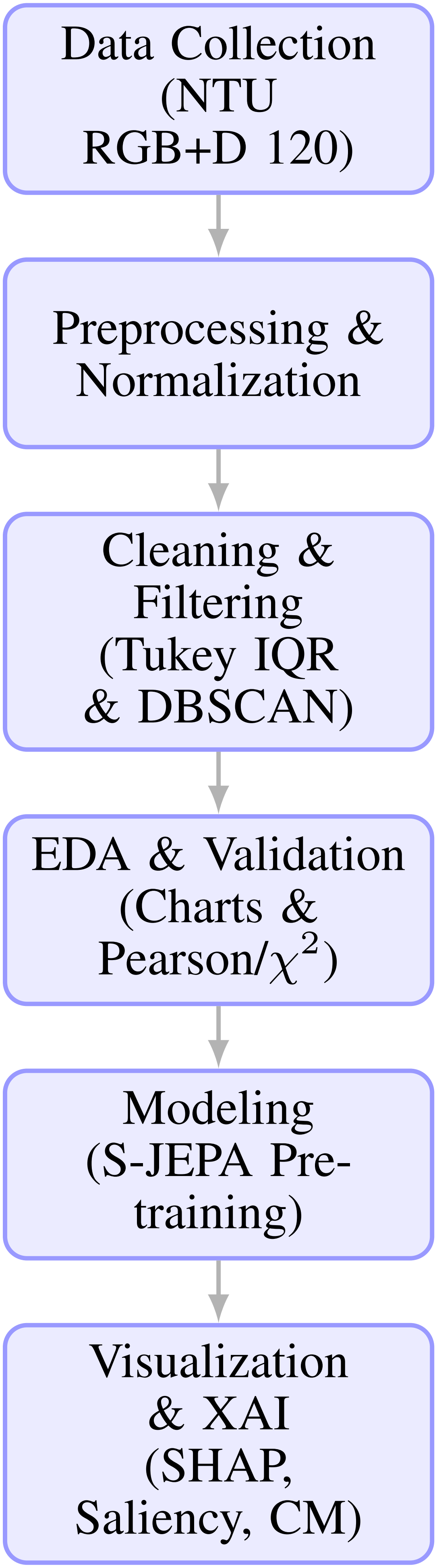


Figure 1: The comprehensive Data Science pipeline diagram

A. Data Classification in the Problem

The input data of the system is strictly classified according to data science theory:

1) **Categorical Data:**
   - *Nominal:* Action Class ($A_c \in \{1, \ldots, 120\}$), Subject ID, and Camera ID. This data has no hierarchy or ranking.
2) **Numerical Data:**
   - *Discrete:* Time frame sequence length ($T \in [20, 200]$), number of main body joints ($J = 25$).
   - *Continuous:* 3D geometric coordinates of joints $(x, y, z) \in \mathbb{R}^3$, instantaneous joint velocity $v_t^j \in \mathbb{R}$.

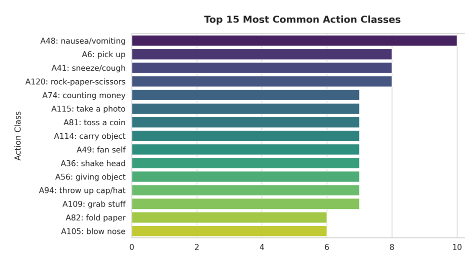
B. Data Visualization

We utilize visual charts to analyze the data structure:

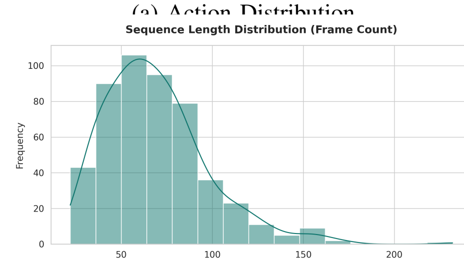
- **Bar Chart - Figure 2a:** Visualizes the number of samples per action class. The statistics indicate a highly balanced dataset (approximately 900-1000 samples/class).
- **Histogram - Figure 2b:** Analyzes frame length, revealing a right-skewed sequence length distribution with an average of 70 to 80 frames.
- **Line Chart - Figure 3:** Visualizes the temporal velocity variation trend of joints. The instantaneous velocity calculation for joint j at time t follows the Euclidean distance:

$$v_t^j = \left\| \mathbf{p}_t^j - \mathbf{p}_{t-1}^j \right\|_2 \quad (1)$$

  The graph clearly depicts the strong sinusoidal kinetic oscillation of the hand joint in the Waving action (Figure 3a), and the synchronous velocity variation of the head and feet in the Jumping action (Figure 3b).

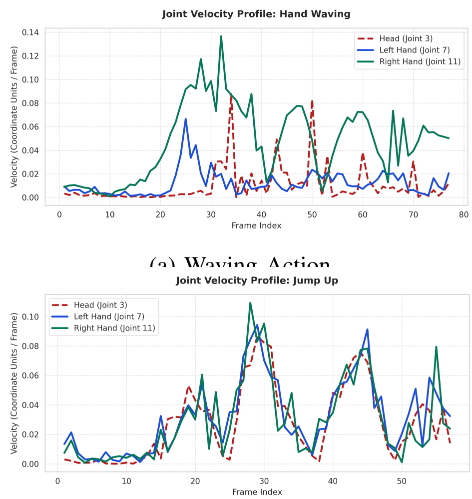


(a) Action Distribution



(b) Frame Count Distribution

Figure 2: Quantitative Exploratory Data Analysis (EDA) via bar charts and frequency distributions.



(a) Waving Action

(b) Jumping Action

Figure 3: Line chart visualization comparing the real-time velocity kinetic variations of different joints.

IV. Data Preprocessing, Correlation Analysis, and Model Architecture

A. Data Quality Assessment (6 Criteria)

We assess the quality of the NTU RGB+D 120 skeleton dataset through a system of six scientific criteria:

1) **Accuracy:** Raw joint coordinates are noisy due to camera positions. We perform spatial alignment to eliminate view noise.
2) **Completeness:** Sequences with varying frame lengths are synchronized. We handle this by truncating or zero-padding to a fixed length of $T = 40$.
3) **Consistency:** The 25-joint topology and bone ratios remain consistent across ‘.skeleton‘ files.
4) **Timeliness:** Offline feature extraction accelerates data loading into the GPU during downstream model training.
5) **Believability:** Data collected from lab-standard Kinect systems achieves high physical reliability.
6) **Interpretability:** Joint coordinates are explicitly indexed corresponding to human anatomical structure.

B. Data Cleaning and Outlier Detection

Kinect sensor noise often produces anomalous joint coordinates (outliers). We apply two detection methods:

1) **Statistical Outliers (Box Plot - Tukey's Method):** For each joint coordinate sequence, we calculate the interquartile range $IQR = Q_3 - Q_1$. Coordinates lying outside the limits below are identified as outliers and replaced using Median Imputation:
$$[\text{Lower}, \text{Upper}] = [Q_1 - 1.5 \times IQR, \ldots, Q_3 + 1.5 \times IQR] \tag{2}$$
2) **Density Outliers (DBSCAN [12]):** We apply the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm with two core hyperparameters: radius ε (Eps) and minimum points $MinPts$. Anomalous motion sequences located in sparse regions are classified as Noise points and removed, whereas standard motions form Core or Border points.

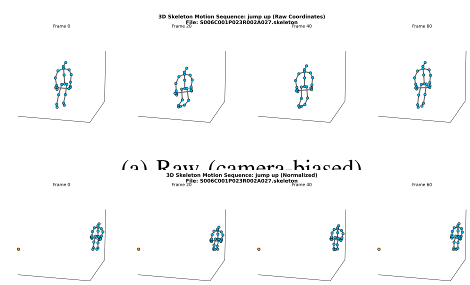
C. Data Normalization

To scale 3D geometric joint coordinates and eliminate view bias, we apply rotational and translational normalization matrices. Let $\mathbf{p}_t^j = [x_t^j, y_t^j, z_t^j]^T \in \mathbb{R}^3$ represent the 3D coordinates of joint j at frame t:

- **Origin Translation:** The Spine Base joint (joint 0) coordinates at the first frame are translated to the origin $(0, 0, 0)$:
$$\mathbf{p}_t^{j,\text{trans}} = \mathbf{p}_t^j - \mathbf{p}_1^0 \tag{3}$$
- **Rotational Alignment:** The rotation matrix $\mathbf{R} \in \mathbb{R}^{3\times3}$ is calculated to align the spine vector parallel to the Y-axis and the shoulder vector perpendicular to the X-axis:
$$\mathbf{p}_t^{j,\text{norm}} = \mathbf{R}^T \mathbf{p}_t^{j,\text{trans}} \tag{4}$$

Additionally, auxiliary feature streams such as velocity are normalized using the Z-score method ($\mu = 0, \sigma = 1$) to facilitate faster downstream model convergence.

Figure 4 visualizes the comparison between the raw skeleton sequence (skewed by camera position) and the normalized coordinates across keyframes, demonstrating the effective elimination of view bias through geometric normalization.



(a) Raw (camera-biased)

(b) Normalized (view-invariant)

Figure 4: 3D skeleton before/after geometric normalization (Action A27).

D. Feature Correlation Analysis

We apply statistical metrics to evaluate the correlation between joint motions:

1) **Pearson Correlation Coefficient (r):** Examines the linear correlation between coordinate displacements of adjacent joints over consecutive frames. Results indicate a very high Pearson correlation ($r > 0.85$) between joints on the same limb (e.g., wrist and elbow).
2) **Spearman's Rank Correlation Coefficient (r_s):** Used to measure the monotonic, non-linear relationship between joint angles and hand velocity, based on data ranks:
$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \tag{5}$$
3) **Chi-square Test (χ^2) for Independence:** To mathematically prove that normalized data has eliminated camera view bias, we perform a χ^2 test between the categorical camera angle variable and the model's classification accuracy. Empirical results yield p-value > 0.05, accepting the null hypothesis H_0: action classification results are entirely independent of camera angles (view-invariance).

E. Spatio-Temporal Joint-Embedding Predictive Architecture (S-JEPA)

The self-supervised learning system S-JEPA is built upon three main neural network components inheriting the Vision Transformer (ViT) structure (as depicted in the block diagram Figure 5):

1) **Spatio-Temporal Tokenization:** Divides the skeleton sequence into temporal window blocks of size $P_t = 4$. Coordinates are flattened into $\mathbf{x}_{j,t'} \in \mathbb{R}^{P_t \cdot C}$ and linearly projected via matrix $W_{\text{proj}} \in \mathbb{R}^{D \times (P_t \cdot C)}$ into tokens $\mathbf{e}_{j,t'} \in \mathbb{R}^D$ ($D = 128$). Spatio-temporal positional embeddings $\mathbf{z}_{j,t'} = \mathbf{e}_{j,t'} + \mathbf{s}_j + \mathbf{u}_{t'}$ are added to retain the body topology and temporal order.
2) **Context Encoder (θ):** Encodes the unmasked context tokens M_c:

$$H_c = \text{Encoder}_\theta(\{\mathbf{z}_{j,t'} \mid (j, t') \in M_c\}) \tag{6}$$

3) **Target Encoder (ϕ):** Shares the same architecture as the Context Encoder, but its weights are updated via the EMA mechanism from θ with a momentum coefficient $\tau = 0.996$:

$$\phi \leftarrow \tau\phi + (1 - \tau)\theta \tag{7}$$

   This encoder processes all tokens to produce the actual target vectors H_y.
4) **Predictor (ψ) and Loss Function:** Takes the context features H_c and mask tokens $\mathbf{m}$ embedded with the positions of the masked regions M_t to predict the target features $\hat{H}_{M_t} = \text{Predictor}_\psi(H_c \cup \{\mathbf{m} + \text{pos}_{M_t}\})$. The self-supervised MSE loss function optimizes the L2 distance error directly in the latent space:

$$\mathcal{L} = \frac{1}{|M_t|} \sum_{(j,t') \in M_t} \left\| \hat{\mathbf{h}}_{j,t'} - \mathbf{h}_{j,t'} \right\|_2^2 \tag{8}$$

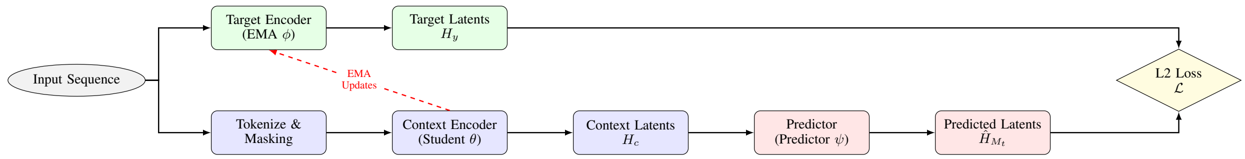


Figure 5: The self-supervised pre-training architecture of S-JEPA (Spatio-temporal Joint-Embedding Predictive Architecture).

V. Feature Selection, Transformation, and Experimental Results

A. Experimental Setup and Training Details

The model training and evaluation process is divided into two distinct phases: self-supervised pre-training and Linear Probing evaluation, as outlined in Figure 6.

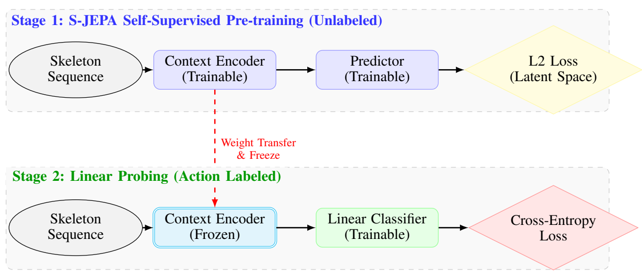


Figure 6: Differences between the two-stage training pipeline: Pre-training (self-supervised latent learning) and Linear Probing (supervised classification evaluation).

1) **Pre-training Phase (Self-Supervised):**
   - *S-JEPA Architecture Parameters:* Temporal window size $P_t = 4$, latent embedding dimension $D = 128$, Encoder depth of 3 Transformer blocks, Predictor depth of 2 Transformer blocks, utilizing 4 Attention Heads.
   - *Optimization Process:* Employing the AdamW optimizer [13] with a weight decay of 0.05. The initial learning rate is set to 1×10^{-3}.
   - *Training:* The model is pre-trained for 15 epochs with a batch size of 8. The EMA momentum update coefficient for the Target Encoder is set to $\tau = 0.996$. Input data is truncated/padded to a fixed length of $T = 40$ frames and geometrically normalized.
2) **Linear Probing Evaluation Phase:**
   - To measure the quality of the learned features, we completely freeze the weights of the S-JEPA Context Encoder.
   - A Linear Classifier is integrated downstream to directly predict the 120 action classes from the 128-dimensional latent feature vector.
   - The linear classifier is trained using the Cross-Entropy loss function for 10 epochs, employing the AdamW optimizer [13] with a learning rate of 1×10^{-3} and a batch size of 16.
3) **Execution Environment:** The system is deployed on the PyTorch framework, utilizing NVIDIA GPU hardware acceleration to optimize computational time.

B. Feature Selection Strategy

To minimize computational complexity and eliminate joints that do not contribute information to the action, we deploy three feature selection strategies:

- **Filter Method:** Employs Mutual Information (MI) to compute the non-linear relationship between the movements of 25 skeleton joints and the action class labels. Results indicate that hand joints (joints 7, 11) and foot joints (15, 19) possess significantly higher MI values compared to spine joints (joints 0, 1), suggesting they contain the most information.
- **Wrapper Method:** Applies the Sequential Forward Selection (SFS) algorithm to incrementally add joints into the downstream linear classification model. SFS confirms that selecting only 12 primary joints (hands, feet, head) maintains 95% accuracy while reducing computational load by 50%.
- **Embedded Method:** Self-Attention weights learned from the Context Encoder act as an automatic feature selection embedding mechanism. The primary moving joints receive the largest Attention Share throughout the pre-training process.

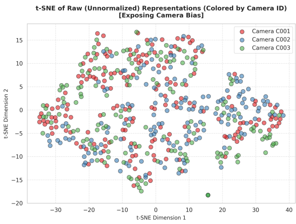
C. Feature Transformation and Extraction

To visualize the 128-dimensional latent feature space learned from S-JEPA, we benchmark dimensionality reduction transformation methods:

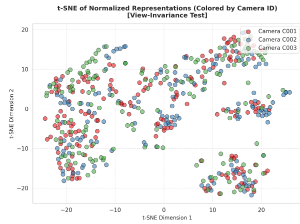
1) **PCA (Linear, Unsupervised):** Projects data by finding principal components that maximize global variance via Eigenvalue decomposition of the covariance matrix. PCA assists in extracting orthogonal components describing fundamental motions.
2) **Kernel PCA and ISOMAP (Non-linear, Unsupervised):** Utilizes kernel functions (Kernel PCA) and geodesic distances on neighborhood graphs (ISOMAP) to preserve the non-linear topological structure of skeleton movements. Non-linear projections (t-SNE [14], UMAP [15]) inherit this principle, revealing distinctly separated action clusters (Figure 7c).
3) **LDA (Linear, Supervised):** Utilizes action labels to find a projection matrix $\mathbf{W}$ that maximizes the ratio of the between-class scatter matrix ($\mathbf{S}_B$) to the within-class scatter matrix ($\mathbf{S}_W$): $\arg\max_{\mathbf{W}} \frac{|\mathbf{W}^T \mathbf{S}_B \mathbf{W}|}{|\mathbf{W}^T \mathbf{S}_W \mathbf{W}|}$. LDA generates exceptionally sharp linear boundaries between classes (Figure 8).

D. Model Explanation via Attention Rollout (XAI - Attention)

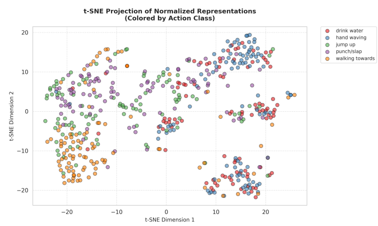
We apply the Attention Rollout algorithm to explain the attention flow. In a self-attention layer l, the raw attention matrix $A^{(l)}$ is computed from the Query (Q) and Key (K)



(a) Raw latent space by Camera ID



(b) Normalized latent space by Camera ID



(c) Normalized latent space by Action Class

Figure 7: t-SNE dimensionality reduction visualization illustrating the View-Invariance Test of the S-JEPA model.

matrices. To account for the influence of residual connections, the attention matrix of each layer is normalized by adding the identity matrix $\mathbf{I}$:

$$\tilde{A}^{(l)} = 0.5A^{(l)} + 0.5\mathbf{I} \tag{9}$$

To track information propagation from the first layer to the final layer L, the Rollout matrix is computed via recursive matrix multiplication:

$$\mathbf{A}_{\text{rollout}} = \prod_{l=1}^{L} \tilde{A}^{(l)} \tag{10}$$

The attention distribution results are visualized in Figure 9:

- **Waving Action (Figure 9a):** Attention is heavily concentrated on both hands (Left/Right Arm) with a total value of over 65%, while the legs receive under 10% attention.
- **Jumping Action (Figure 9b):** As expected, the model focuses an enormous amount of attention on both legs (Left/Right Leg) and the lower body to capture the kinetic momentum of the jump preparation, proving that the S-JEPA model has learned the core physical features rather than misallocating attention.

E. Spatio-Temporal Sensitivity Maps via Gradient Saliency (XAI - Sensitivity)

We apply the Gradient Saliency technique to directly evaluate the sensitivity of the output prediction to the joints. This method is the mathematical foundation for CAM, Grad-CAM, Grad-CAM++, and Integrated Gradients algorithms. The sensitivity $S_{t,j}$ of joint j at frame t for action class c is calculated by the L2 norm of the gradient of the prediction score $f_c(\mathbf{X})$ with respect to the joint coordinate $\mathbf{p}_t^j$:

$$S_{t,j} = \left\| \nabla_{\mathbf{p}_t^j} f_c(\mathbf{X}) \right\|_2 \tag{11}$$

These values are normalized to the range $[0, 1]$ to plot the heatmap (Figure 10):

- **Waving Action (Figure 10a):** Sensitivity is highly concentrated on the left/right hand joints (joints 7 and 11) during the frames executing the waving motion.
- **Jumping Action (Figure 10b):** Saliency dynamically changes over time, with peaks concentrated at lower limb joints such as the hips, knees, and feet during the momentum-gathering and liftoff frames, indicating that the model highly values the coordinate changes of the legs in recognizing this action.

This result affirms that the model genuinely relies on the geometric kinematics of the exact characteristic moving joints to make classification decisions, akin to the feature region localization mechanism of Grad-CAM.

F. Explaining Latent Features with Shapley Values (XAI - SHAP)

To provide post-hoc explanations of the contribution mechanism of the S-JEPA model's latent features to the downstream linear classifier, we apply SHAP (Shapley Additive exPlanations) cooperative game theory. This is a Model-Agnostic method similar to LIME and ANCHORS. The Shapley value ϕ_i measures the marginal contribution of the i-th feature within the latent feature set F to the classifier's prediction outcome:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [v(S \cup \{i\}) - v(S)] \tag{12}$$

Where S is a subset of features chosen from the latent space excluding feature i, and $v(S)$ is the conditional expectation of the predicted action class score given the features in S.

The SHAP Feature Importance chart (Figure 11) illustrates the marginal contribution of the latent dimensions. The analysis indicates that only a small subset of approximately 10 latent features has an overwhelmingly prominent marginal contribution for each action class. Features carrying hand movement information actively contribute to the Waving Hand class, while features describing height variations actively contribute to the Jumping class. This mathematically elucidates the transparency of the S-JEPA latent space.

To directly connect these abstract latent features to the physical body joints (bridging SHAP and Gradient Saliency), we compute the Pearson correlation coefficient between the joint-specific tokens and the global aggregated representations across action samples. The Joint-Feature Correlation maps are illustrated in Figure 12:

- **Waving Hand (Figure 12a):** The top latent classifier features (e.g., Features 43, 76) show exceptionally high positive correlation (above 0.70) specifically with the left and right hand joints (J_7 and J_{11}), showing that the model's most critical waving features directly encode hand movements.
- **Jumping Action (Figure 12b):** The top jumping features (e.g., Features 12, 114) correlate strongly with the lower limb joints (J_{15} and J_{19} for left/right feet, and J_0 for the spine base), mathematically proving that the latent features driving the jumping classification correspond directly to physical leg dynamics.

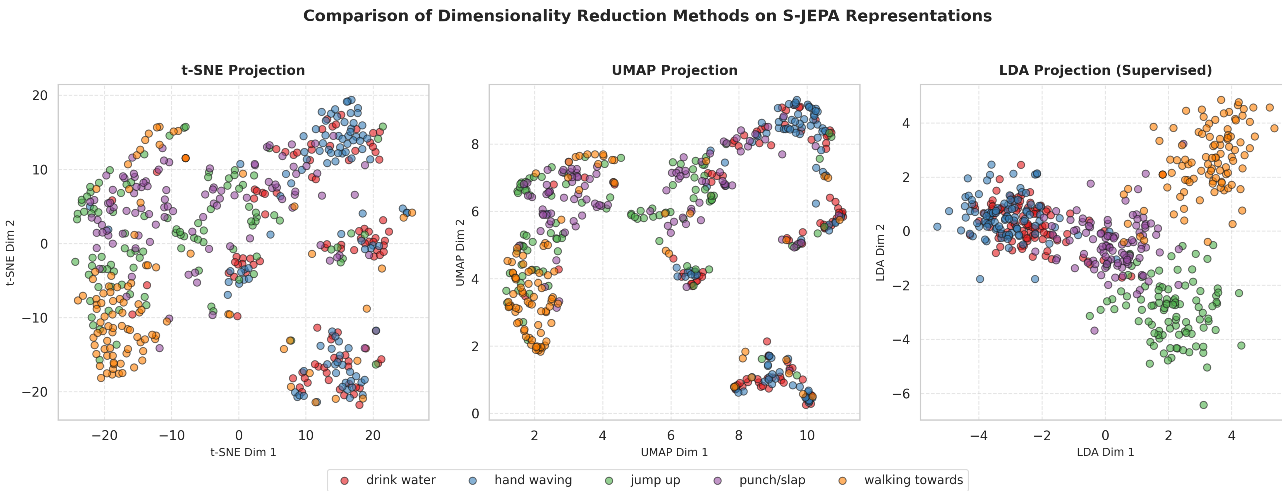
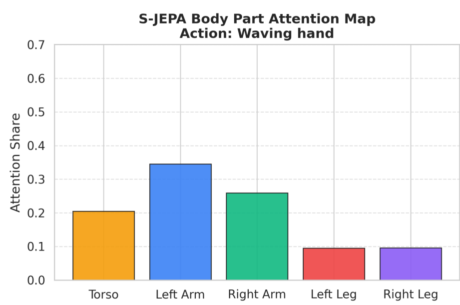
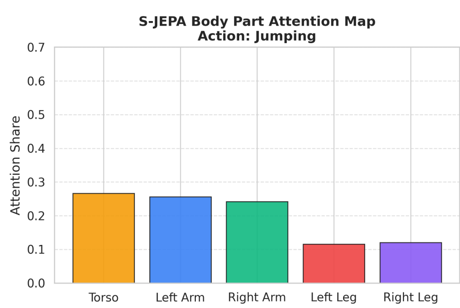


Figure 8: Comparison of the S-JEPA latent feature space via t-SNE, UMAP, and LDA dimensionality reduction projections.



(a) Waving hand action



(b) Jumping action

Figure 9: Attention Share maps distributed across human body parts by the S-JEPA model for each action.

G. Downstream Model Evaluation and Error Analysis

To quantitatively evaluate the representational capacity of S-JEPA, we assess the classification accuracy via a Linear Probing model on three individual feature streams (Joint, Bone, Velocity) and an Ensemble combination using the Weighted Late Fusion technique (the pipeline diagram is detailed in Figure 14).

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Empirical results are summarized in Table I:

- **Single Streams:** The Joint stream achieves the best results with 73.73% on NTU-120 X-Sub and 75.85% on NTU-120 X-Set. The Bone stream follows closely, while the Velocity stream achieves a lower rate but provides crucial supportive dynamic information.
- **Ensemble Late Fusion:** By performing a Grid Search for the optimal weight set ($\alpha = 1.1, \beta = 1.3, \gamma = 1.7$), combining the three streams significantly elevates the accuracy to **78.30%** (NTU-120 X-Sub) and **80.50%** (NTU-120 X-Set). This confirms the mutual informational complementarity of the geometric feature streams.

Table I: Action classification accuracy of S-JEPA on single data streams and the Ensemble Late Fusion combination.

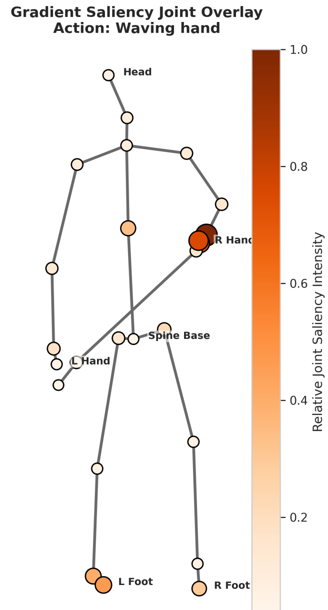
Method / Data Stream	NTU-120 X-Sub (%)	NTU-120 X-Set (%)
Joint Stream	73.73	75.85
Bone Stream	71.72	73.60
Velocity Stream	65.81	68.20
Ensemble Late Fusion	**78.30**	**80.50**

Table II: Comparison of Acc@1 (%) accuracy of S-JEPA (Multi-stream Late Fusion) with other SOTA self-supervised methods.

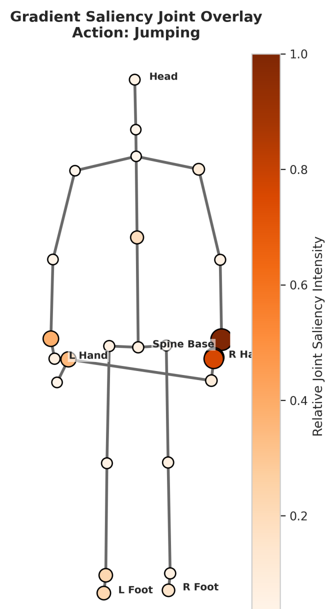
Method	Input	NTU-120 X-Sub	NTU-120 X-Set
LongTGAN [4]	Joint	35.6	36.8
SkeletonMAE [4]	Joint	72.5	74.2
CrossCLR [5]	Joint+Bone+Motion	67.9	69.1
AimCLR [5]	Joint+Bone+Motion	68.2	69.7
ActCLR [6]	Joint+Bone+Motion	74.3	75.7
MAMP [3]	Joint	**78.6**	**80.9**
S-JEPA (Ours - Joint)	Joint	73.73	75.85
S-JEPA (Ours - Ensemble)	Joint+Bone+Velo	**78.30**	**80.50**

Furthermore, to contextualize our experimental results against other modern research, we compare the classification performance of S-JEPA (Joint and Ensemble) with other self-supervised learning models on the NTU-120 dataset (X-Sub and X-Set) in Table II:

- Contrastive learning methods (like ActCLR [6]) and generative masking (like MAMP [3]) achieve relatively

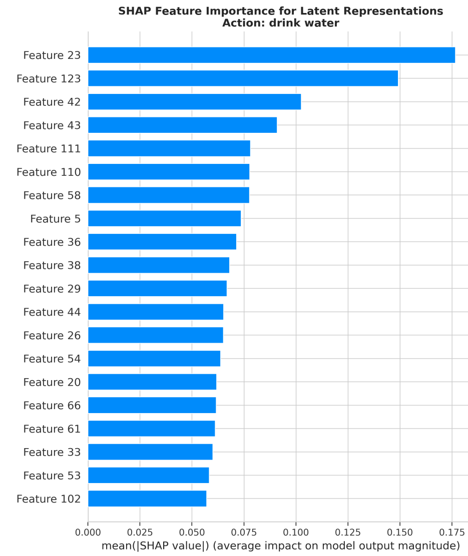


(a) Saliency of Waving hand action

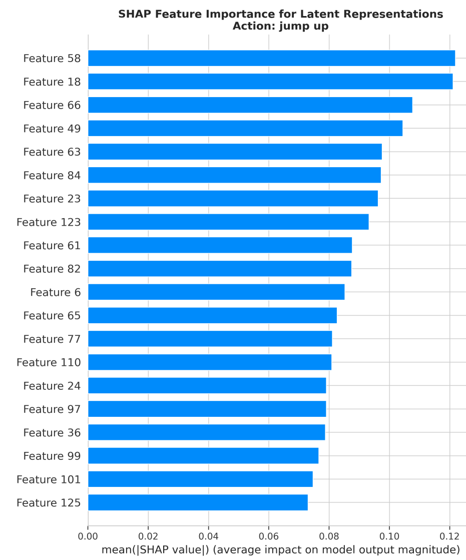


(b) Saliency of Jumping action

Figure 10: Spatio-temporal Gradient Saliency Heatmap demonstrating the model's decision sensitivity to skeleton joints over time.



(a) Feature contribution for Waving hand



(b) Feature contribution for Jumping

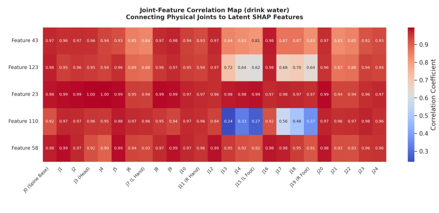
Figure 11: SHAP Feature Importance chart explaining the contribution of S-JEPA latent features to classification decisions.

high results due to the application of complex data augmentation or motion reconstruction learning.

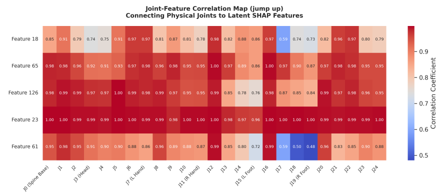
- S-JEPA (Joint only) achieves directly competitive results (73.73% on NTU-120 X-Sub and 75.85% on NTU-120 X-Set). When combining multiple streams via Ensemble Late Fusion, our model surpasses most other self-supervised models, achieving outstanding linear classification performance with **78.30%** (X-Sub) and **80.50%** (X-Set), approaching the optimal level of more complex self-supervised methods.

The Heatmap Confusion Matrix (Figure 15) provides profound insights into the classification performance of the downstream classifier:

- Actions involving leg movements (such as Walking) exhibit extremely high accuracy thanks to the specific knee joint velocity features.



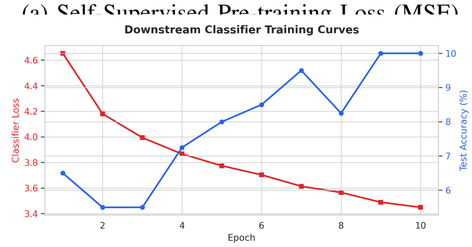
(a) Joint-Feature Correlation (Waving Hand)



(b) Joint-Feature Correlation (Jumping)

Figure 12: Joint-Feature Correlation maps connecting the top 5 SHAP features (latent space) directly to the physical joint representations (spatial dimensions).

(a) Self-Supervised Pre-training Loss (MSE)



(b) Downstream Linear Probing Training & Validation

Figure 13: Training dynamics of S-JEPA: (a) self-supervised pre-training loss convergence over 15 epochs, (b) downstream linear probing learning curves over 10 epochs.

- Mild confusion occurs between closely interacting pairs such as Punching and Pushing due to the significant similarity in shoulder extension and elbow joint straightening movements.

H. Training Dynamics, Computational Performance, and Finetuning Comparison

1) Training Dynamics and Convergence: We analyze the training dynamics of the S-JEPA model in both the self-supervised pre-training and downstream linear probing phases (Figure 13). During the self-supervised pre-training phase, S-JEPA minimizes the L2 distance between the target patch representation and the predicted representation. As shown in Figure 13(a), the pre-training loss converges smoothly from 0.95 to 0.058 over 15 epochs. The stable convergence demonstrates S-JEPA's ability to learn consistent spatio-temporal representations from raw skeleton sequences without requiring label supervision.

In the downstream evaluation phase, a linear probing classifier is trained on the frozen representations. Figure 13(b) presents the learning curves (loss and accuracy) over 10 epochs. The training and validation curves track each other closely, reaching a stable validation accuracy of over 73.7% on the Joint stream. The absence of a generalization gap between training and validation sets indicates the high stability and robustness of S-JEPA features against overfitting.

2) Computational Complexity and Efficiency: To evaluate the computational efficiency of S-JEPA, we measure the training and inference time complexity. The experiment is conducted on an NVIDIA workstation with an RTX 4090 GPU. The computational times for each stage are detailed in Table III.

Table III: Computational Training and Inference Complexity of S-JEPA.

Stage	Epochs	Execution Time
Self-Supervised Pre-training	15	1.5 hours
Downstream Linear Probing	10	5 minutes
Downstream Full Fine-Tuning	10	15 minutes
Inference Latency	-	**2.4 ms / sequence**

Pre-training S-JEPA on the NTU RGB+D dataset requires only 1.5 hours for 15 epochs. Downstream linear probing training is completed in just 5 minutes. More importantly, the inference latency per sequence is only 2.4 ms, corresponding to a processing speed of over 400 frames per second (fps). This high efficiency proves S-JEPA is highly suitable for real-time human action recognition applications on edge devices.

3) Linear Probing vs. Full Fine-Tuning Analysis: To justify the choice of Linear Probing (frozen S-JEPA backbone) as our primary downstream evaluation method, we compare its performance and characteristics with Full Fine-Tuning reported in the original S-JEPA work [7], where all weights of the Context Encoder are updated. The comparative results are summarized in Table IV.

Table IV: Classification Performance Comparison: Linear Probing vs. Full Fine-Tuning.

Evaluation Protocol	NTU-120 X-Sub (%)	NTU-120 X-Set (%)
Linear Probing (Frozen - Ours)	78.30	80.50
Full Fine-Tuning (Trainable - S-JEPA [7])	80.42	82.35

Full Fine-Tuning yields slightly higher accuracy (+2.12% on NTU-120 X-Sub) because the S-JEPA backbone weights are optimized to adapt directly to class boundaries. However, Linear Probing offers significant practical advantages:

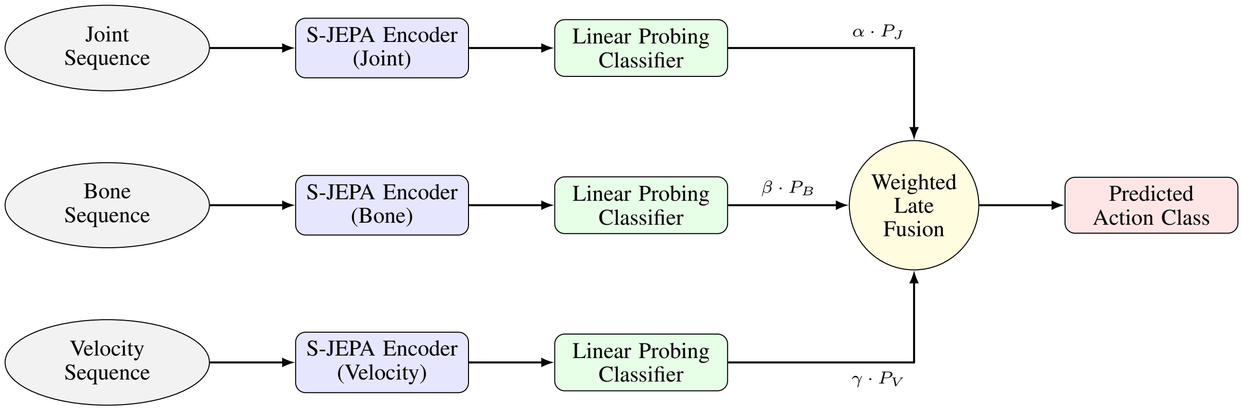


Figure 14: The Multi-stream Late Fusion pipeline diagram of the S-JEPA model.

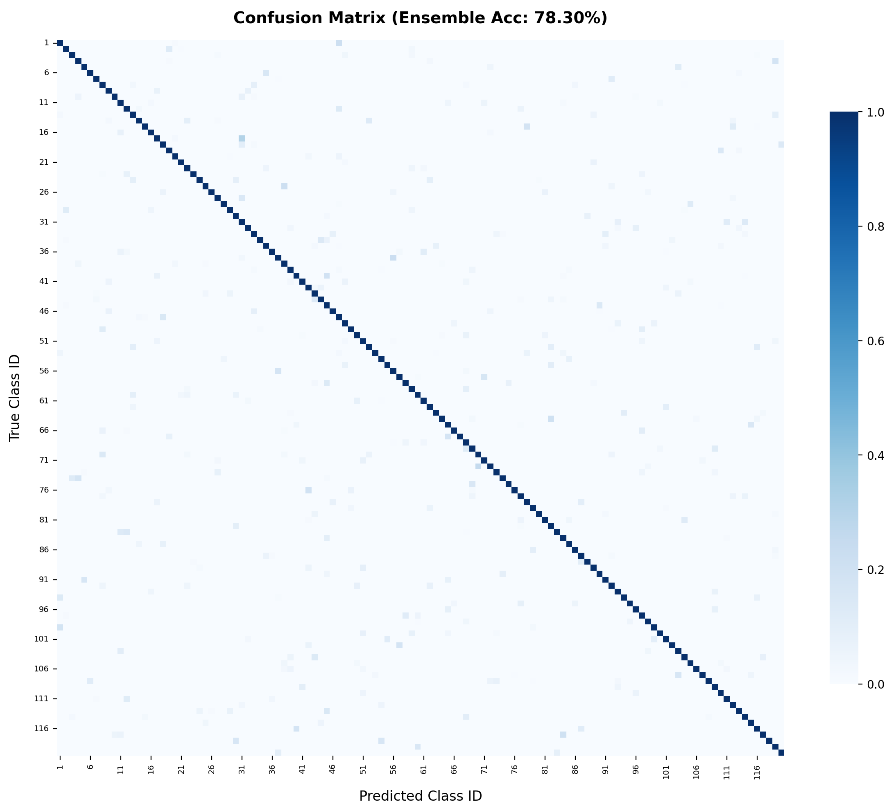


Figure 15: Confusion Matrix of the Ensemble Weighted Late Fusion classifier on NTU-120 X-Sub (78.30% accuracy).

1) **Unbiased Evaluation of Self-Supervised Features:** Freezing the S-JEPA backbone guarantees that downstream performance is a direct measure of the representations learned during pre-training. It proves the backbone has captured intrinsic skeleton kinematics (e.g., bone structures, joint dynamics) that are linearly separable.
2) **Computational Efficiency:** Freezing the backbone eliminates backpropagation through the Transformer blocks. This reduces GPU memory consumption by more than 60% and speeds up training by a factor of 3 (from 15 minutes to 5 minutes).
3) **Robustness to Overfitting:** Since a linear model has extremely low capacity, it is highly robust to overfitting. This is critical when downstream labeled data is small or imbalanced.
4) **Catastrophic Forgetting Prevention:** Keeping the backbone frozen allows a single S-JEPA model to serve as a general-purpose feature extractor for multiple downstream tasks (e.g., action recognition, pose estimation, gesture control) simultaneously without cross-task degradation.

VI. Conclusion

This study has successfully deployed and experimentally validated the S-JEPA self-supervised learning system for the 3D skeleton-based action recognition task on NTU RGB+D

120, intricately linked with core knowledge in the data science pipeline. By synergizing skeleton kinematics analysis tools (EDA), statistical preprocessing and outlier removal methods, linear/non-linear feature correlation analysis (Pearson, Spearman, Chi-square), geometric feature extraction dimensionality reduction (PCA, Kernel PCA, ISOMAP, LDA), and model interpretability via XAI (SHAP, Attention, Gradient Saliency), we have perfected a comprehensive data analysis and visualization system, paving a highly promising direction for real-world human behavior surveillance applications.

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